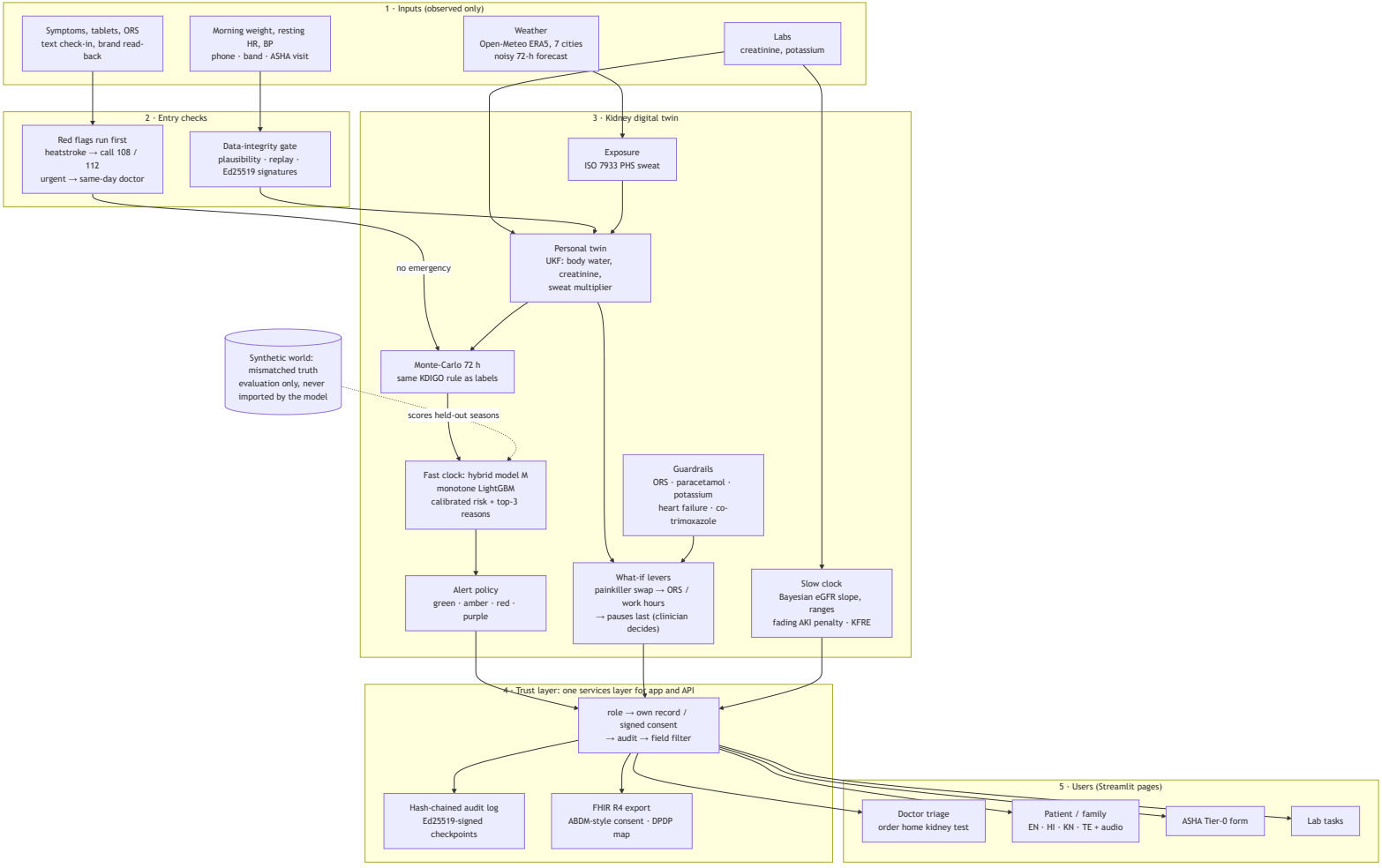

1 · Inputs (observed only)
Symptoms, tablets, ORS
text check-in, brand read-back
Morning weight, resting HR, BP
phone · band · ASHA visit
Weather
Open-Meteo ERA5, 7 cities
noisy 72-h forecast
Labs
creatinine, potassium
2 · Entry checks
Red flags run first
heatstroke → call 108 / 112
urgent → same-day doctor
Data-integrity gate
plausibility · replay · Ed25519 signatures
3 · Kidney digital twin
Exposure
ISO 7933 PHS sweat
Personal twin
UKF: body water, creatinine, sweat multiplier
no emergency
Synthetic world:
mismatched truth
evaluation only, never imported by the model
Monte-Carlo 72 h
same KDIGO rule as labels
scores held-out seasons
Fast clock: hybrid model M
monotone LightGBM
calibrated risk + top-3 reasons
Guardrails
ORS · paracetamol · potassium
heart failure · co-trimoxazole
Alert policy
green · amber · red · purple
What-if levers
painkiller swap → ORS / work hours
→ pauses last (clinician decides)
Slow clock
Bayesian eGFR slope, ranges
fading AKI penalty · KFRE
4 · Trust layer: one services layer for app and API
role → own record / signed consent
→ audit → field filter
Hash-chained audit log
Ed25519-signed checkpoints
FHIR R4 export
ABDM-style consent · DPDP map
5 · Users (Streamlit pages)
Doctor triage
order home kidney test
Patient / family
EN · HI · KN · TE + audio
ASHA Tier-0 form
Lab tasks